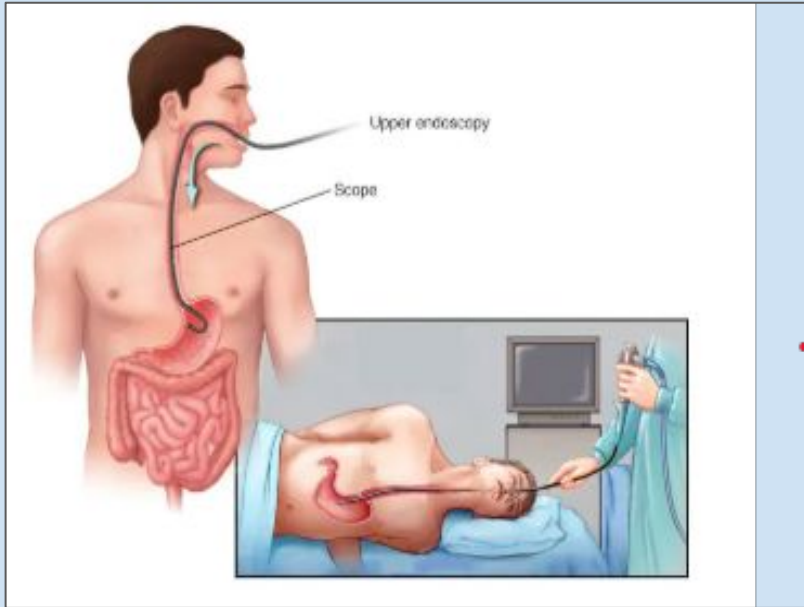


Diffusion Modeling for Speckle Generation in Multimode Fibers

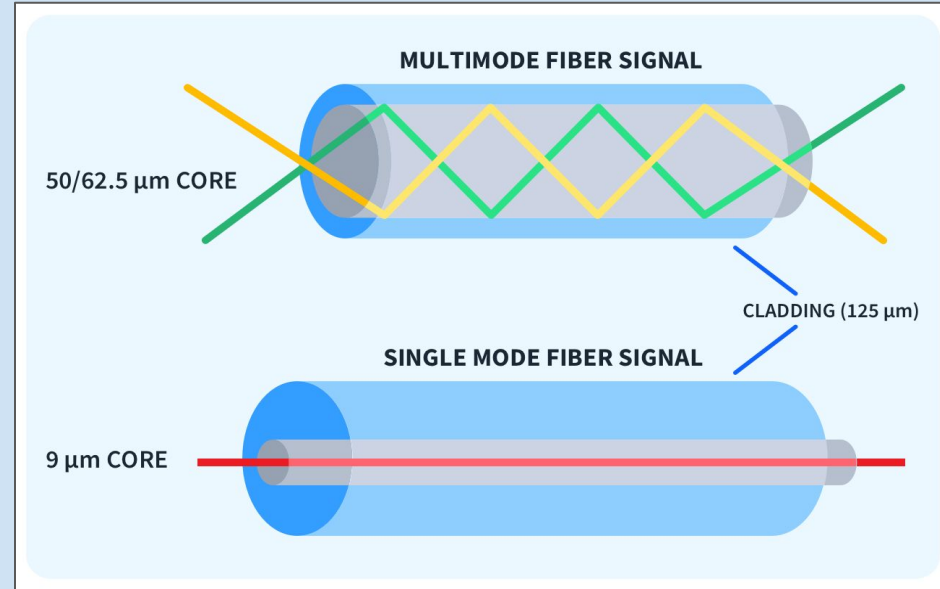
P.G Subhash
Supervised by Dr.Pengfei Fan -2025
Queen Mary University of London.

How it started? Endoscopy?

A significant focus is on developing ultra-thin, flexible endoscopes for high-resolution imaging deep within biological tissues without causing substantial damage.



<https://gastroclinic.com.sg/blog/how-do-endoscopy-and-gastroscopy-differ/>



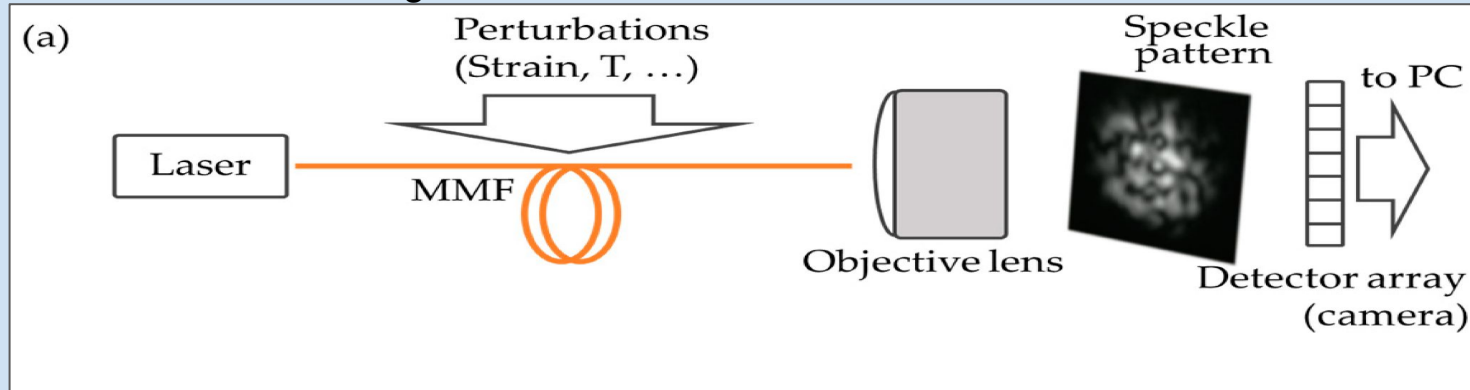
<https://images.ctfassets.net/aoyx73g9h2pg/2akZ34C0SwKh3IRZg3u0M5/bbdf30bbe3d7f6939a78d50df925a28/Single-Mode-vs-Multimode-Fiber-Diagram.jpg>

The Power of Multimode Fibers

- **A New Era of Minimally Invasive Technology:** Multimode Fibers (MMFs) are tiny and flexible, offering a revolutionary way to perform imaging inside the body without extensive surgery. Think of them as a key to next-generation endoscopes.
- **Advanced Biomedical Applications:**
 - **Deep-Brain Imaging:** Visualizing neurons and cell structures in real time.
 - **Targeted Diagnosis:** Using light to identify specific biological tissues, like lipids, for clinical diagnosis.
 - **Better Medical Scopes:** Giving doctors a clear, close-up look inside the body for surgery and diagnosis.

Speckle Images

1. **Speckle Formation Mechanism:** In multimode fibers (MMFs), speckle patterns arise from the complex interference of multiple propagating optical modes, leading to intricate intensity distributions at the output.
2. **Deterministic Behavior:** Although the resulting speckle patterns appear random, they are deterministic and repeatable within a specific temporal window, assuming stable input conditions and fiber configuration.



Challenges

The optical field at the distal end appears as a noise-like or random speckle pattern with randomly distributed bright and dark spots.

This is an "unfavorable speckle pattern" that typically requires calculating the fiber transmission matrix in advance.

Limitations of Traditional Wavefront Shaping (WFS) and Transmission Matrix (TM) Methods:

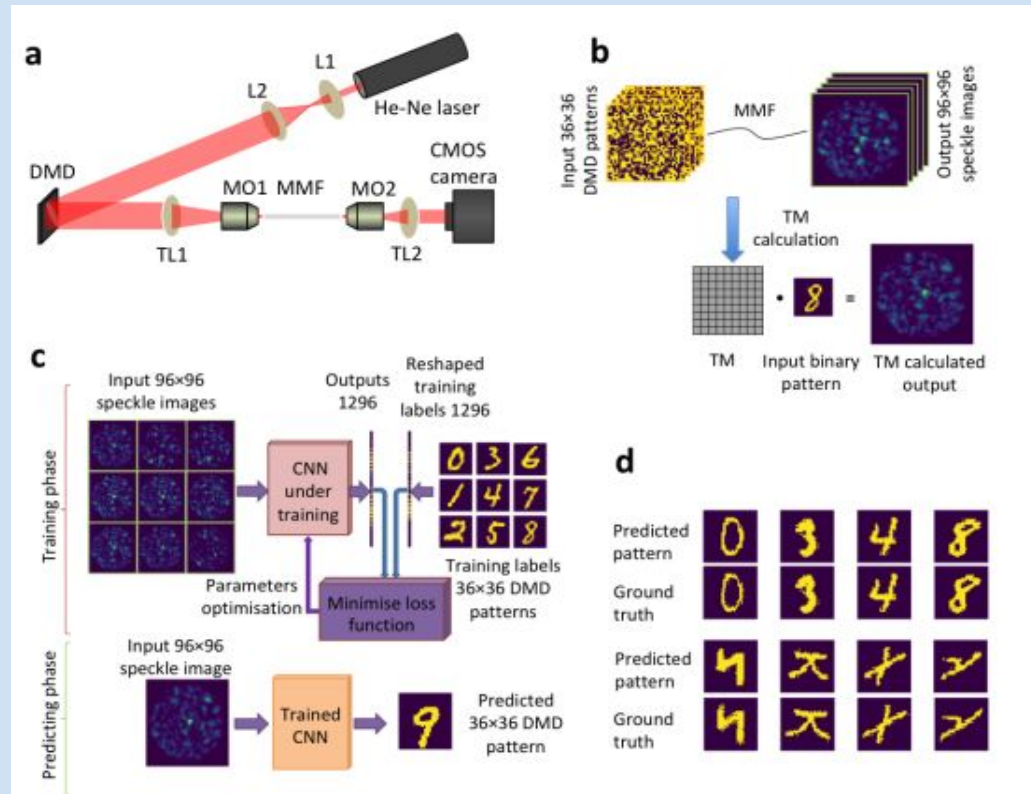
- **Sensitivity to Perturbations:** Factors such as **fiber bending, mechanical stress, temperature fluctuations, and wavelength drift** can completely scramble optical information and cause the speckle patterns to shift and decorrelate over time.
- **Requirement for Phase Information**
- **Computational Complexity and Time Consumption**
- **Lack of Generalization**

Machine Learning Addresses These Challenges

- Robustness & generalization
- Intensity-only measurements
- High speed & real-time capabilities
- Handling nonlinearity
- Computational imaging & other applications

How it was addressed?

- ML enables control over MMFs, turning them from chaotic light scramblers into predictable, programmable optical systems.
- Structured inputs like MNIST digits and A–Z letters are used to modulate light entering the fiber.
- Light is shaped using a Spatial Light Modulator (SLM) or Digital Micromirror Device (DMD), not physical images.
- Speckle patterns and labels from datasets to train deep networks for classification, reconstruction, or output control.



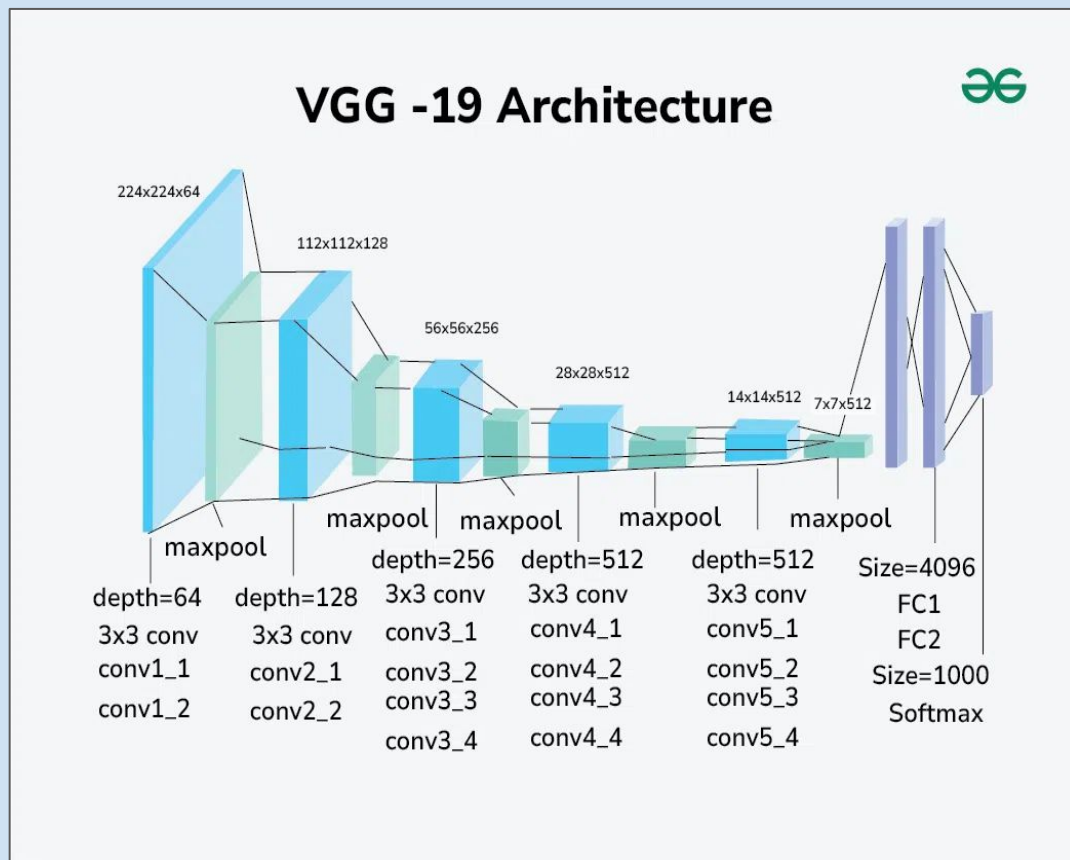
Purpose and Research Question

- How can we evaluate the assigned pattern to the new speckle?
- How can modern generative engines help in creating the speckle reflection image for the new pattern?

CNN's Background

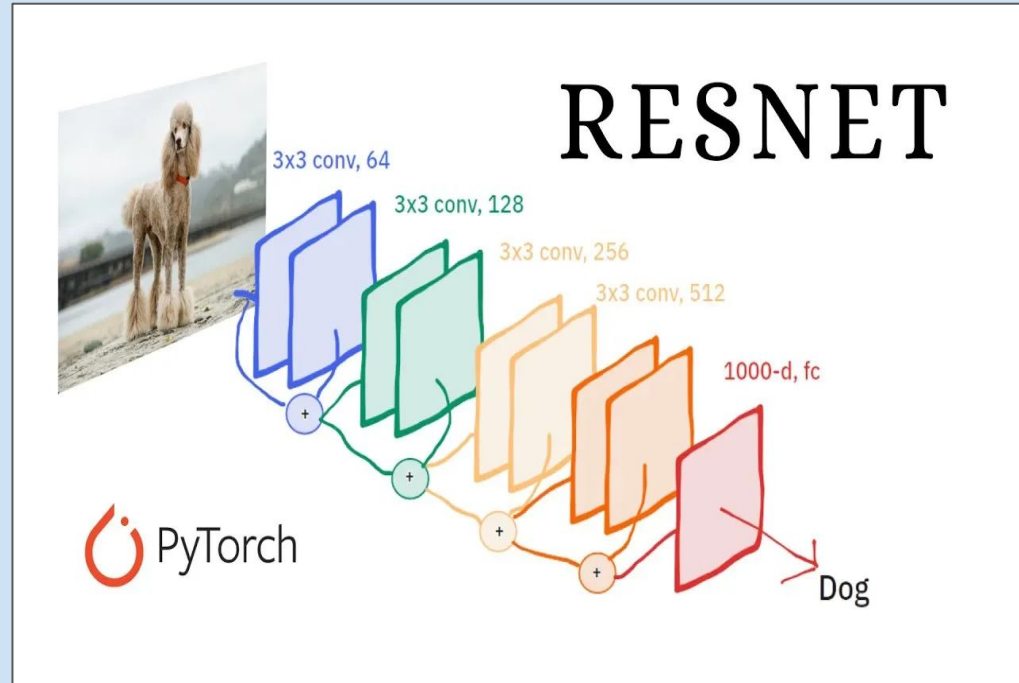
A typical convolutional network layer has three stages:

- convolutions for linear activations.
- a detector stage with a nonlinear activation function.
- and a final pooling stage.



ResNets: A Breakthrough in Deep Learning

- Introduced by He et al. (2015)
- Used **residual (skip) connections** to bypass layers
- Solved **degradation problem** in very deep networks



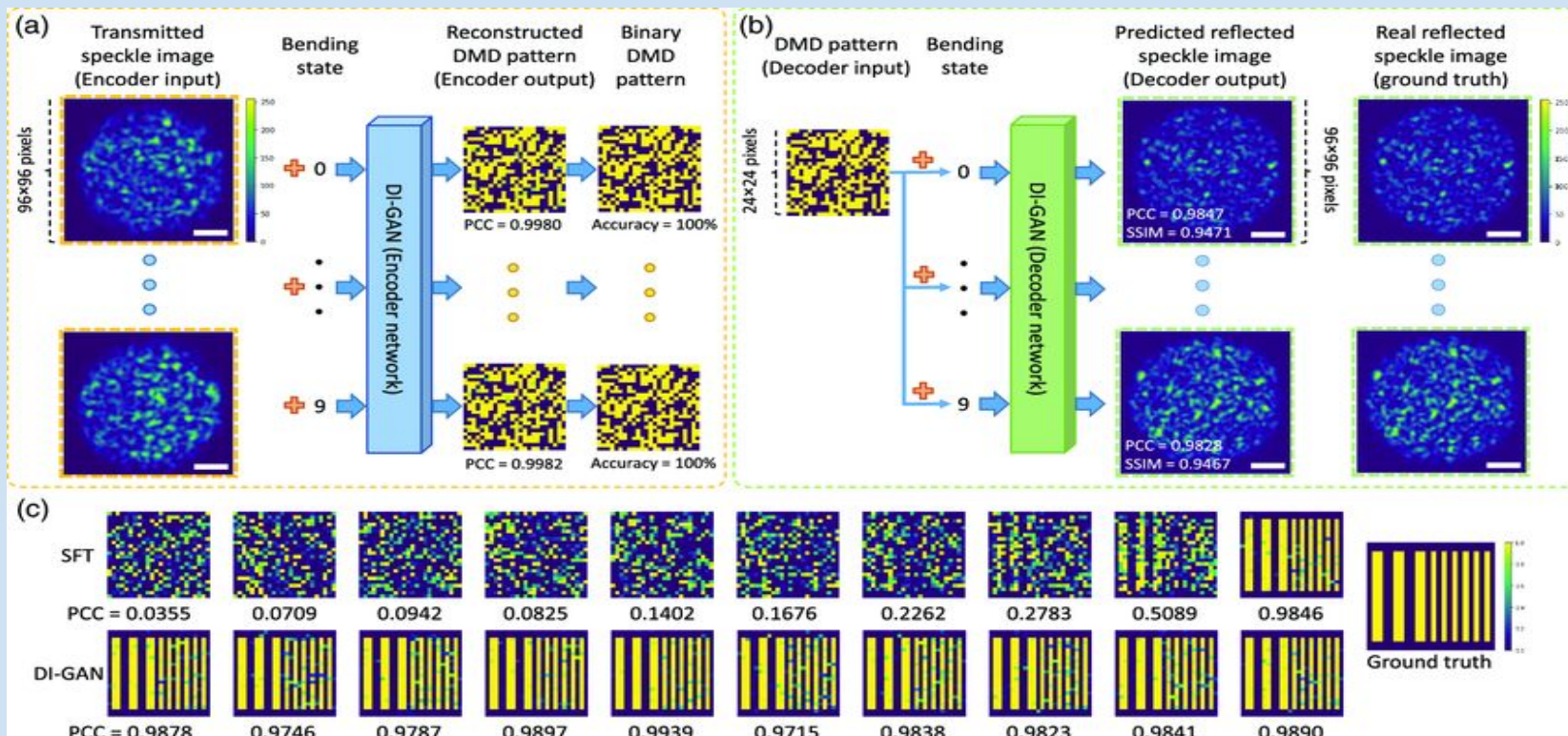
Application in Speckle Synthesis

Deep learning models, particularly CNNs, have been used to predict input or output patterns in MMFs.

Various CNN and deep learning architectures have been successfully applied:

- **U-Net**
- **DI-GAN**
- **VGG-type Networks**
- **Dense Neural Networks (SHL-DNN)**
- **Autoencoder (AE-SNN)**
- **Mode Transformer Network (MTNet)**

Although **DI-GAN** is not a **traditional generative model** like a GAN or VAE that generates images from noise or latent vectors, it **performs generative tasks** in the context of **multimode fiber (MMF) speckle pattern modeling** by **generating or reconstructing optical patterns** under complex physical conditions.



Current Limitations of DNNs & DI-GAN

- **Poor Generalization**

- DNNs trained on specific datasets (e.g., digits) may fail on unseen or complex image types (e.g., natural scenes).
- Performance drops outside the training data distribution.

- **Overfitting Risks**

- Complex models + limited data → prone to overfitting → weak generalization.

- **Fixed Architecture (DI-GAN)**

- Cannot dynamically adapt to new tasks or data distributions.
- Requires **dynamic expansion & loss function redesign** for better data synthesis.

- **Inconsistent Speckle Generation**

- Deep learning can generate speckles, but **output quality & uniformity vary**, often due to modulator or network limitations.

Why Diffusion?

1. High-Fidelity Speckle Generation

Gap: Deep learning often produces inconsistent or low-fidelity speckles.

Diffusion Solution:

Generates visually coherent, realistic speckle patterns.

Improves fidelity and uniformity → benefits downstream applications (e.g., focusing, reconstruction).

2. Learning Complex Forward Mapping

Gap: Traditional models fail to represent non-linear MMF behavior.

Diffusion Solution:

Learns the input → output mapping statistically from data.

Captures modal crosstalk, scrambling, and noise without explicit physical modeling.

Acts as a generic approximator for MMF forward propagation.

Introduction to Diffusion Models

Diffusion models represent a powerful class of generative models that have demonstrated state-of-the-art performance in synthesizing high-fidelity images.

High-fidelity generation: Diffusion models achieve state-of-the-art performance in synthesizing realistic images.

Forward diffusion: Gradually adds noise to data in multiple steps, corrupting the original structure.

Reverse diffusion (learned): The model learns to denoise step by step, reconstructing data from noise.

Stable training: Avoids issues like mode collapse and adversarial instability common in GANs.

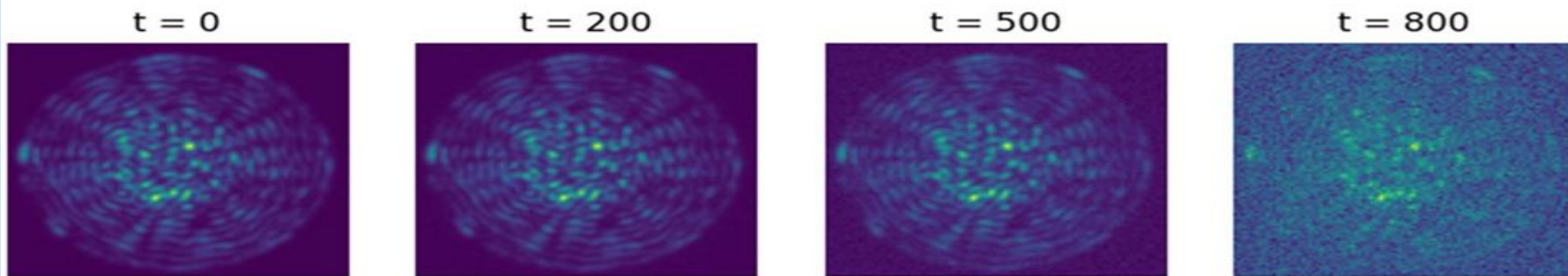
The Denoising Diffusion Probabilistic Model (DDPM)

It consists of two primary processes: a fixed forward diffusion process and a learned reverse denoising process.

Forward Diffusion Process:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I})$$

Progressive Gaussian Noise on Speckle Image

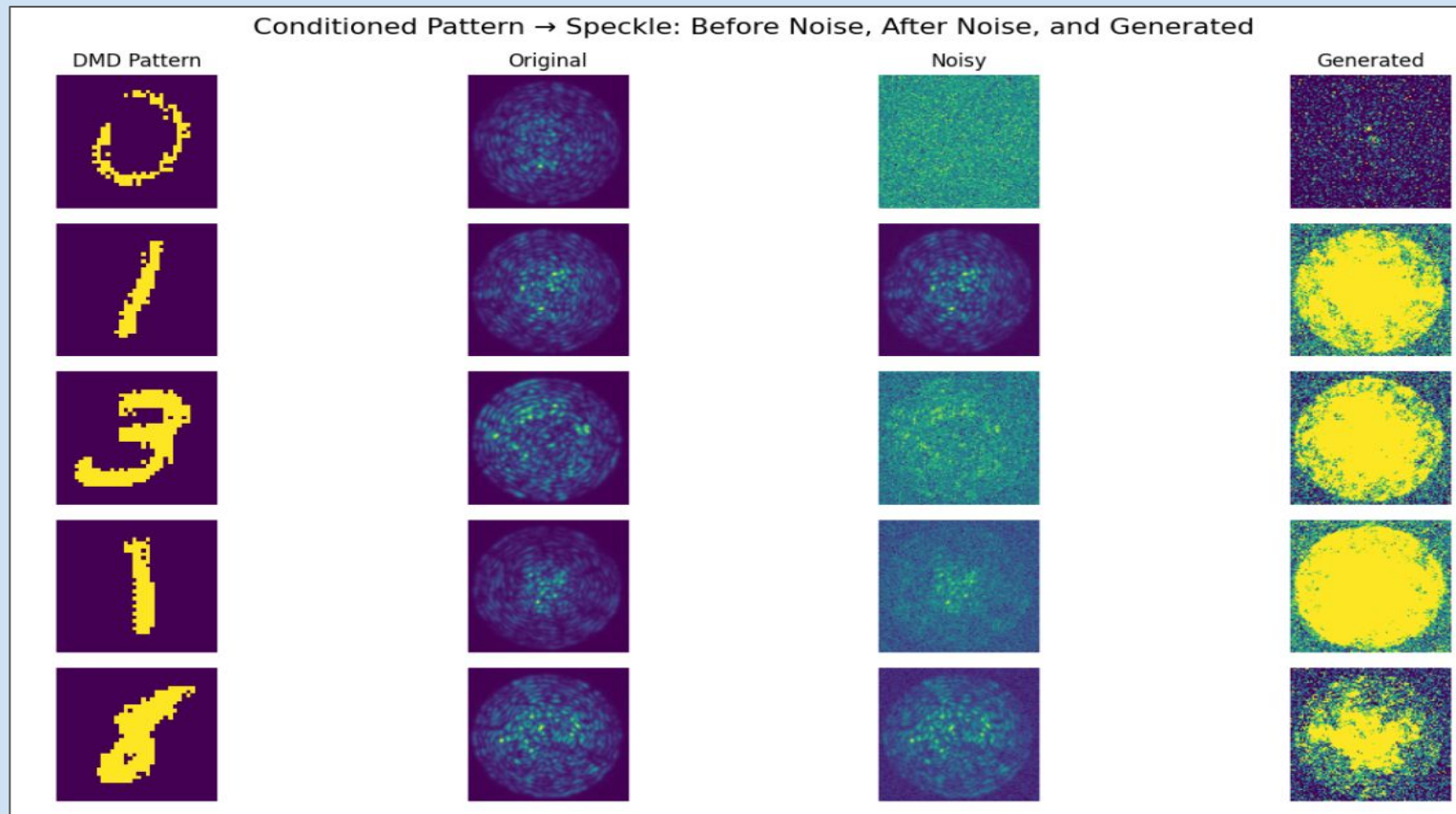


Reverse Process

The reverse process is a learned Markov chain that starts from a pure Gaussian noise vector, $x_T \sim \mathcal{N}(0, I)$, and progressively removes the noise over T steps to generate a clean image. This process is modeled by a neural network, typically a U-Net, which is trained to predict the noise added at each step.

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

Initial failure with conditioned DMD pattern



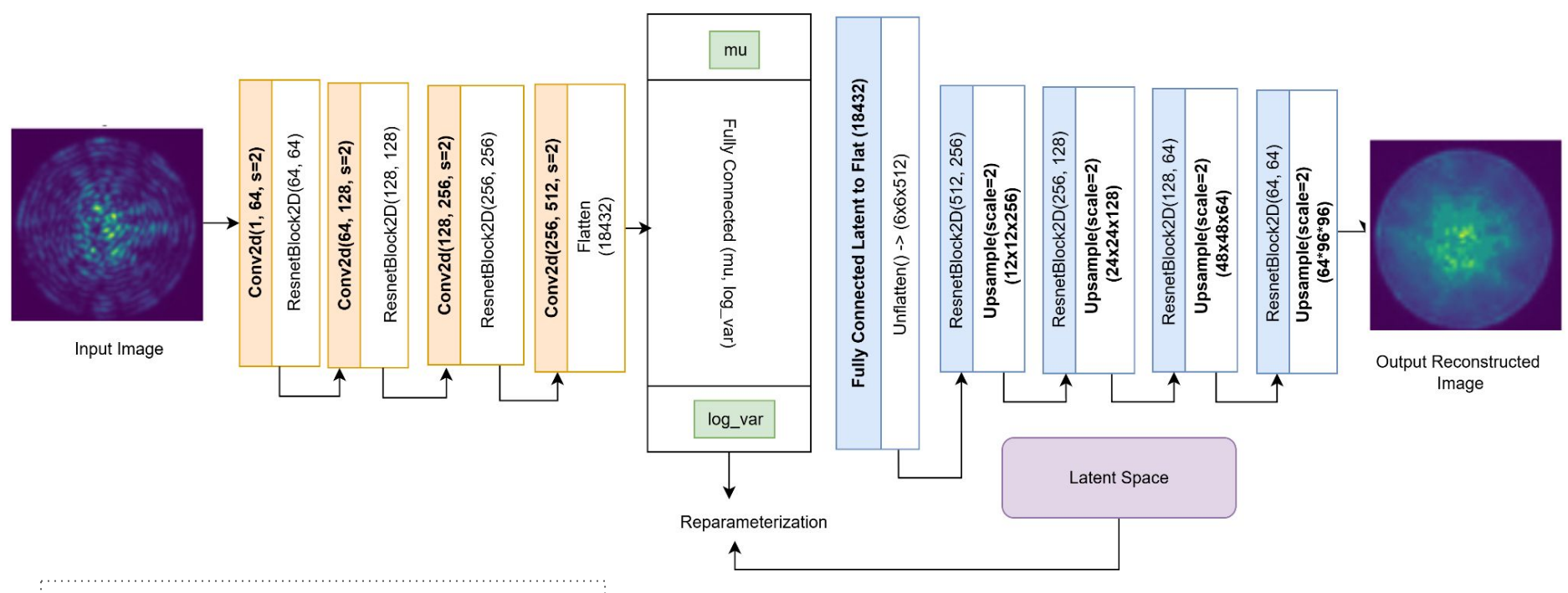
Progress to Stable Diffusion Network

- The initial approach trained the model directly on images, conditioning on corresponding pattern images.
- This baseline architecture, comprising only the standard forward and reverse (denoising) processes, was not satisfactory and model was unable to generate.

VAE's:

For each latent attribute of training data, VAEs encode two different latent vectors: a vector of means, “ μ ,” and a vector of standard deviations, “ σ .”

ResNet VAE Architecture



$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}$$

The encoder consisted of four convolutional layers using ResNet blocks with kernel size 3×3 , stride 2, and SiLU activation function, followed by two fully connected layers producing the mean vector μ and the log-variance vector $\log \sigma^2$.

Resnet VAE loss evaluation - Beta parameter

- $\mathcal{L}_{\text{rec}}$ is the **reconstruction loss**, often MSE:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|_2^2$$

- $\mathcal{L}_{\text{KL}}$ is the **Kullback-Leibler divergence** between the learned latent distribution $q(z|x) \sim \mathcal{N}(\mu, \sigma^2)$ and a standard normal prior $p(z) \sim \mathcal{N}(0, 1)$:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$$

$$\mathcal{L}_{\beta\text{-VAE}} = \mathcal{L}_{\text{rec}} + \beta \cdot \mathcal{L}_{\text{KL}}$$

Why Resnet?

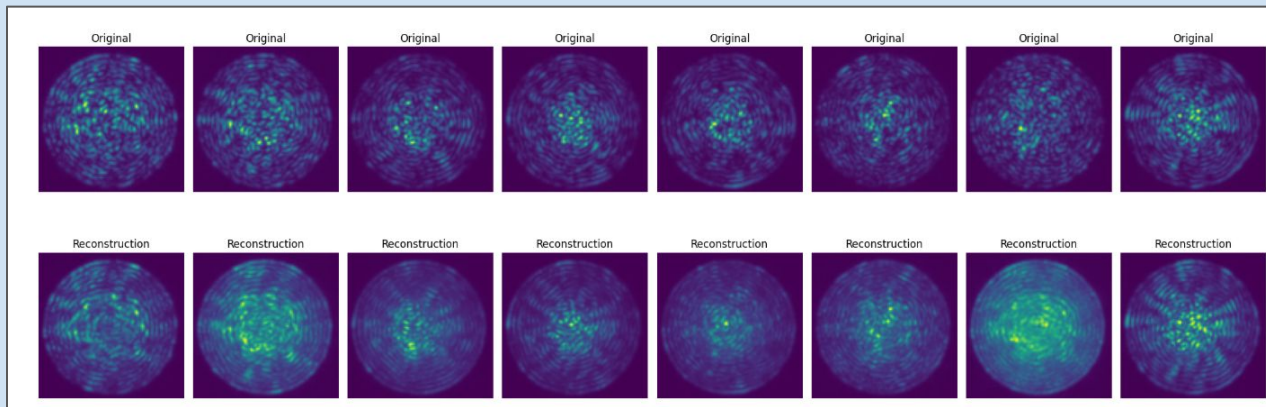
ResNet (Residual Networks) introduce **skip connections**, which add the input of a block directly to its output:

- Avoiding Vanishing Gradients
- Better Feature Extraction
- Stability in Training
- Improved Reconstruction Quality
- Regularization Effect

Table 4.1: Training performance comparison of different VAE architectures.

Model	Latent Dim	Training Time (hrs)	Final Loss
Reconstruction Quality			
Basic CNN VAE	256	1	~0.015
Low to Medium			
ResNet VAE (64→512)	512	2.5	~0.0057
High			
ResNet VAE Wide (96→768)	512	3.5	~0.004
Overly Sharp / Over-constructive			

VAE Reconstructions



$\beta = 1$: Standard VAE

$\beta > 1$: Encourages **disentanglement** in the latent space at the cost of reconstruction accuracy

$\beta < 1$ (rare): Prioritizes reconstruction, less regularization

U-net for Denoising

- **U-Net Backbone** – The model uses a U-shaped architecture with downsampling (encoder), bottleneck, and upsampling (decoder) for image generation from noisy speckle inputs.
- **Conditioning Inputs** – It is conditioned on **diffusion timestep** (via sinusoidal positional embeddings) and **pattern image** (processed by a separate conditioning encoder).
- **Feature Modulation** – Conditioning is applied through **FiLM layers**, which modulate intermediate feature maps, guiding the generation process more effectively.

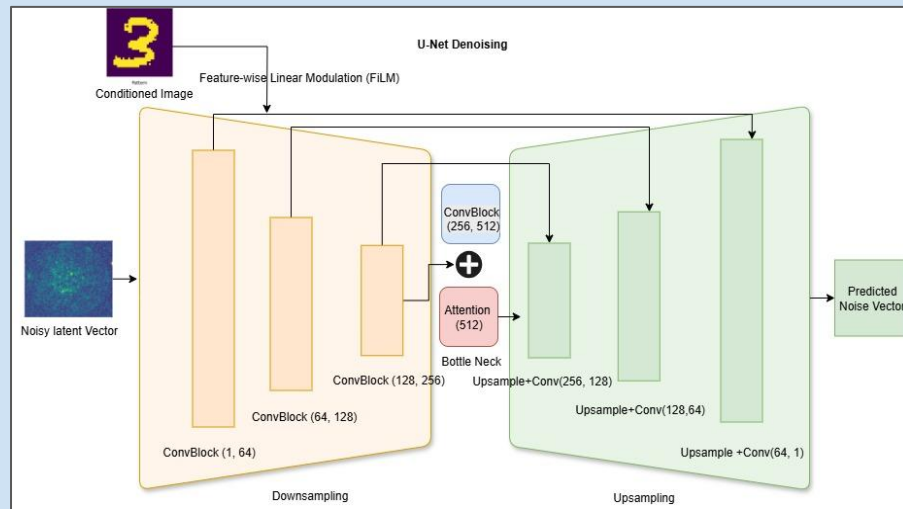
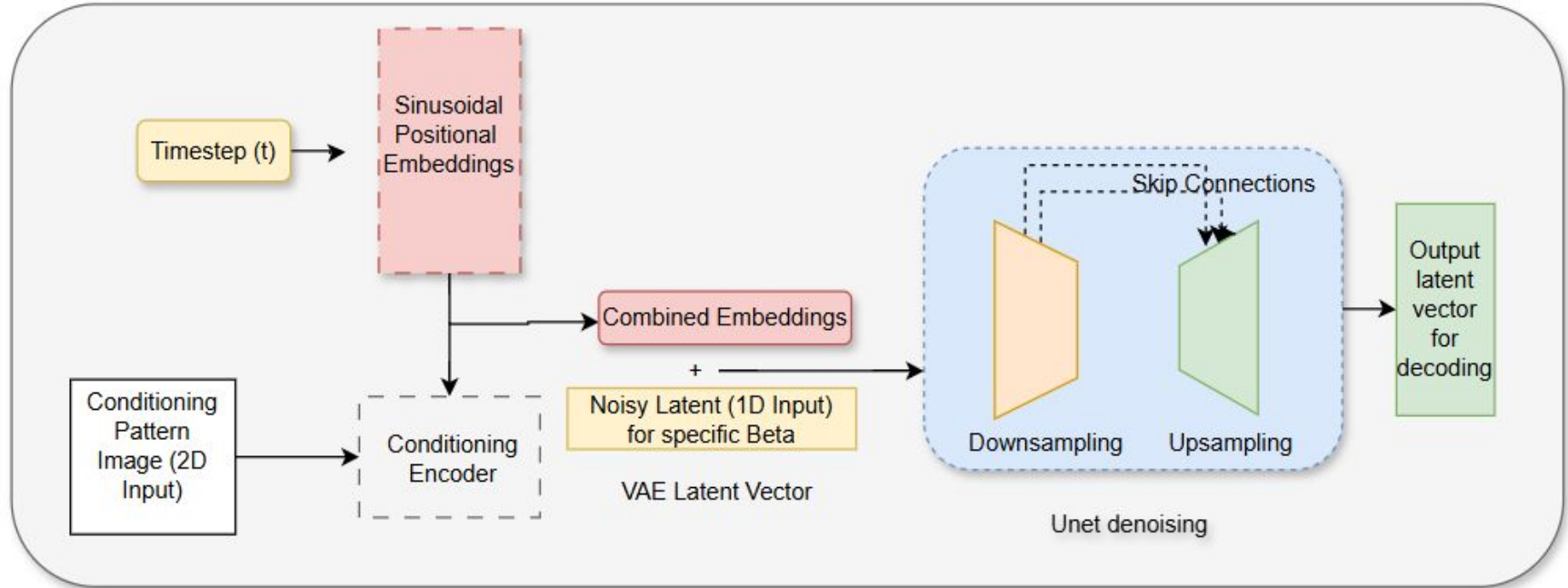


Figure 3.2: **U-Net Architecture for Latent Denoising.** The diagram illustrates the U-Net, which is the core of the latent diffusion model. The network takes a noisy latent vector (here it is shown as image for understanding but in actual its a noisy vector) as input and is conditioned on a pattern image. The downsampling path reduces the feature maps through a series of ConvBlock and pooling. Skip connections transfer feature maps to the upsampling path. The bottleneck includes an Attention block to capture global dependencies. The final output is a predicted noise vector.

Complete diffusion Process



Forward Diffusion

- Start from latent z_0 (VAE encoder output).
- Noise is added over T steps:

$$q(z_t | z_{t-1}) = \mathcal{N}(z_t; \sqrt{1 - \beta_t} z_{t-1}, \beta_t I)$$

Noise Schedule

- Linear variance schedule ($T=1000$)

$$\beta_t \in [\beta_{\text{start}}, \beta_{\text{end}}], \quad \beta_{\text{start}} = 0.0001, \beta_{\text{end}} = 0.02$$

Data Preprocessing

- Each latent vector normalized

$$z \mapsto \frac{z - \mu(z)}{\sigma(z)}$$

Noise Prediction Task

- At each step, model receives noisy latent z_t , timestep embedding t , and conditioning c (pattern image).
- U-Net predicts noise

$$\hat{\epsilon}_{\theta}(z_t, t, c) \approx \epsilon$$

Loss Function

- Train by minimizing **Smooth L1 Loss**

$$L_{\text{diffusion}} = \text{SmoothL1Loss}(\hat{\epsilon}_{\theta}(z_t, t, c), \epsilon)$$

Why Smooth L1 loss?

Mean Squared Error (MSE):

- Penalizes **large errors more heavily** (quadratic growth).
- Sensitive to **outliers** — if one timestep/noisy latent is badly predicted, the loss explodes and can destabilize training.
- Tends to push the model toward **average predictions** → can lead to blurry reconstructions.

L1 loss :

- Penalizes errors **linearly**, making it **more robust to outliers**.
- Encourages **sparser, sharper** predictions compared to MSE.
- However, plain L1 can have unstable gradients near 0.

Smooth L1 loss:

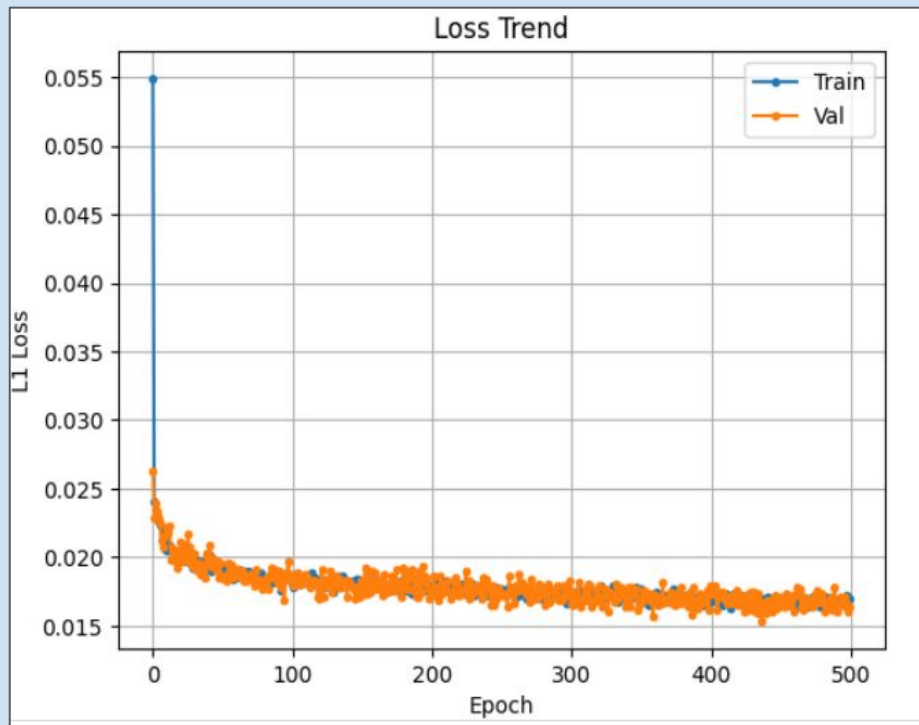
- Behaves like **MSE for small errors** (stable convergence).
- Behaves like **L1 for large errors** (robust to outliers).
- Strikes balance → smoother optimization + sharper reconstructions.

Summary:

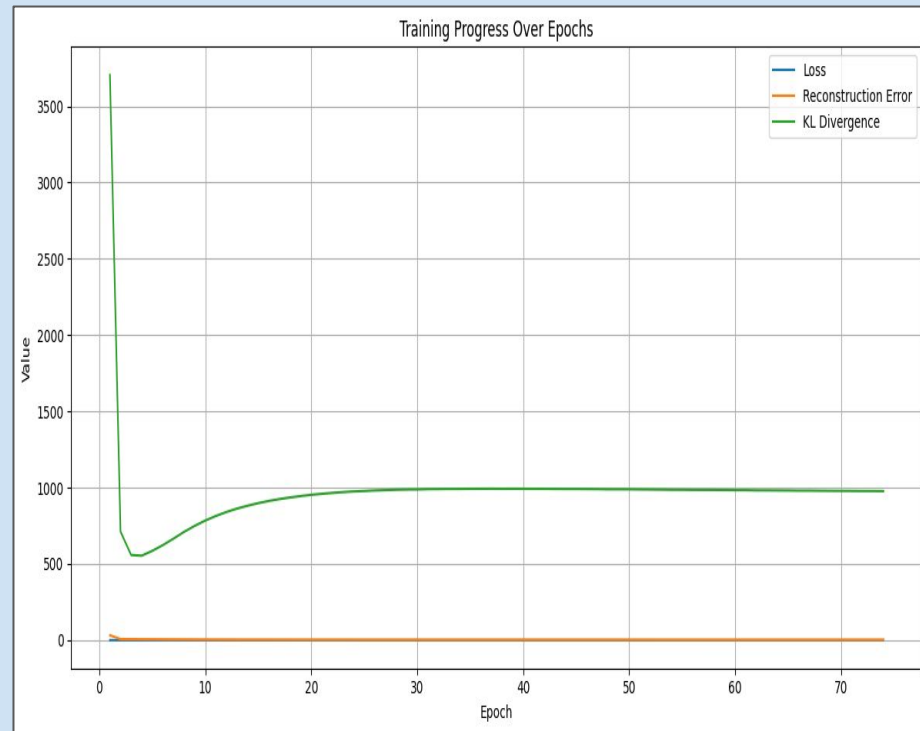
Table 5.1: Summary of U-Net Training Phases and Outcomes

Phase	Experiment	Scheduler Used	Best Validation Loss	Outcome & Insight
1. Baseline	Initial U-Net Training	Standard	~ 0.21	Stuck. Model is functional but has a high loss.
2. Diagnosis	Retrain U-Net on "best reconstruction" VAE latents	ReduceLR-OnPlateau	0.1223	Failure. Sharp VAE reconstructions don't guarantee a good latent space.
3. Breakthrough 1	Retrain U-Net on new, well-balanced ResnetVAE latents	ReduceLR-OnPlateau	0.0814	Success. Better VAE and latent space is the key to improving performance.
4. Breakthrough 2	Retrain U-Net on the same ResnetVAE latents	CosineAnnealing-LR	0.0243	Final Success. A state-of-the-art scheduler can further optimize the training.

Training curves



Training Curve of U-Net Denoising



Training Curve of VAE

Limitations

Table 5.2: Structural Similarity Index (SSIM) Scores for Different Models

Model	Image 1	Image 2	Image 3	Image 4	Image 5	Average SSIM
Proposed Model	0.923	0.918	0.920	0.915	0.924	0.920
Model B	0.81	0.79	0.80	0.82	0.78	0.80
Model C	0.61	0.59	0.60	0.62	0.58	0.60

What we couldn't cover (time constraints)

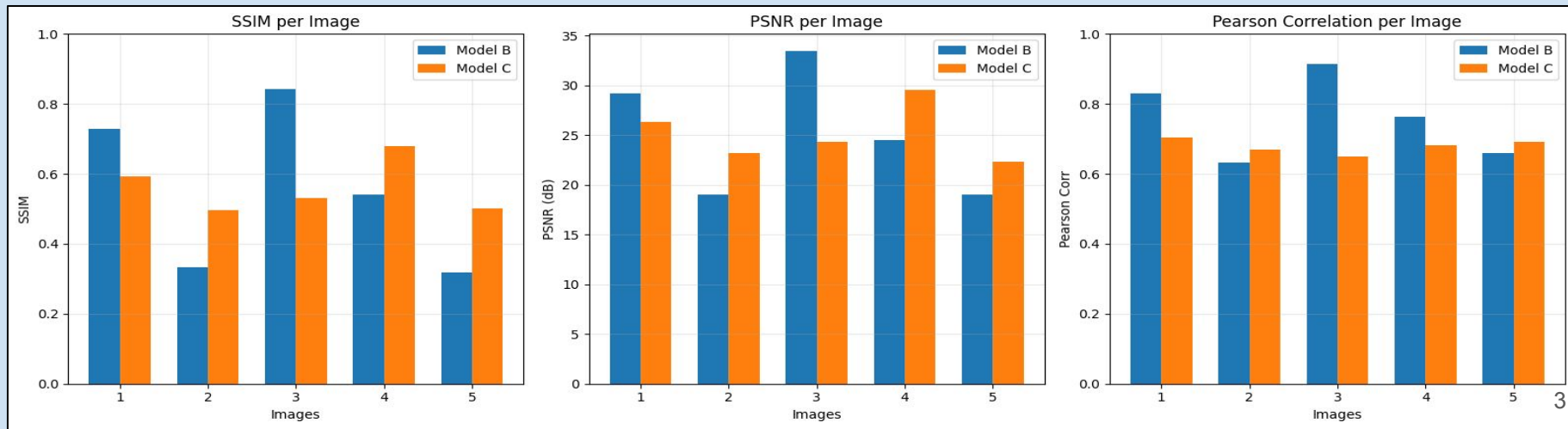
- **GAN baselines not defended:** I did not run/defend a full set of GAN comparisons within the timeframe.
- **Bending-state generalization not defended:** I did not complete an evaluation across different fiber bending states.

Why this is reasonable

- **Scope focus:** Prioritized VAE-latent diffusion (training stability + fidelity) over maintaining multiple adversarial baselines.
- **Compute/time limits:** Full GAN tuning and multi-state data collection were beyond feasible resources for this cycle.

Comparisons & Proposed Work:

- **LoRA/Q-LoRA for bending states:** Freeze the base U-Net/latent-diffusion model; attach low-rank adapters to attention/conv blocks; fine-tune per bending state with a *small* calibration set.
- **Protocol:** For each bend state b : collect $N \sim 100 - 500N$, latent-speckle pairs $\rightarrow$ **train LoRA/Q-LoRA for 10–50 epochs** $\rightarrow$ **evaluate on held-out data**.
- **Metrics:** PSNR/SSIM on reconstructions, speckle contrast & correlation vs. ground truth, task metrics (focusing efficiency), plus FID/KID on generated speckle realism.
- **Ablations:** (i) zero-shot base vs LoRA-adapted; (ii) rank rrr (e.g., 4/8/16); (iii) Q-LoRA (4-bit) for memory-constrained runs.



Experimental & Computational Details

Dataset & DMD Patterns

- A. Total samples: 38,000 image pairs
- B. Input: Binary 36×36 DMD patterns (from MNIST)
- C. Output: 96×96 speckle images
- D. Train/Validation split: 80% / 20%
- E. Geometry used: G1 (bending 1)

Diffusion Model Settings

- a. Diffusion steps (T) & noise schedule: 1000 timesteps
- b. Latent dimension: 512
- c. Loss functions: L1 Loss

Hardware & Framework

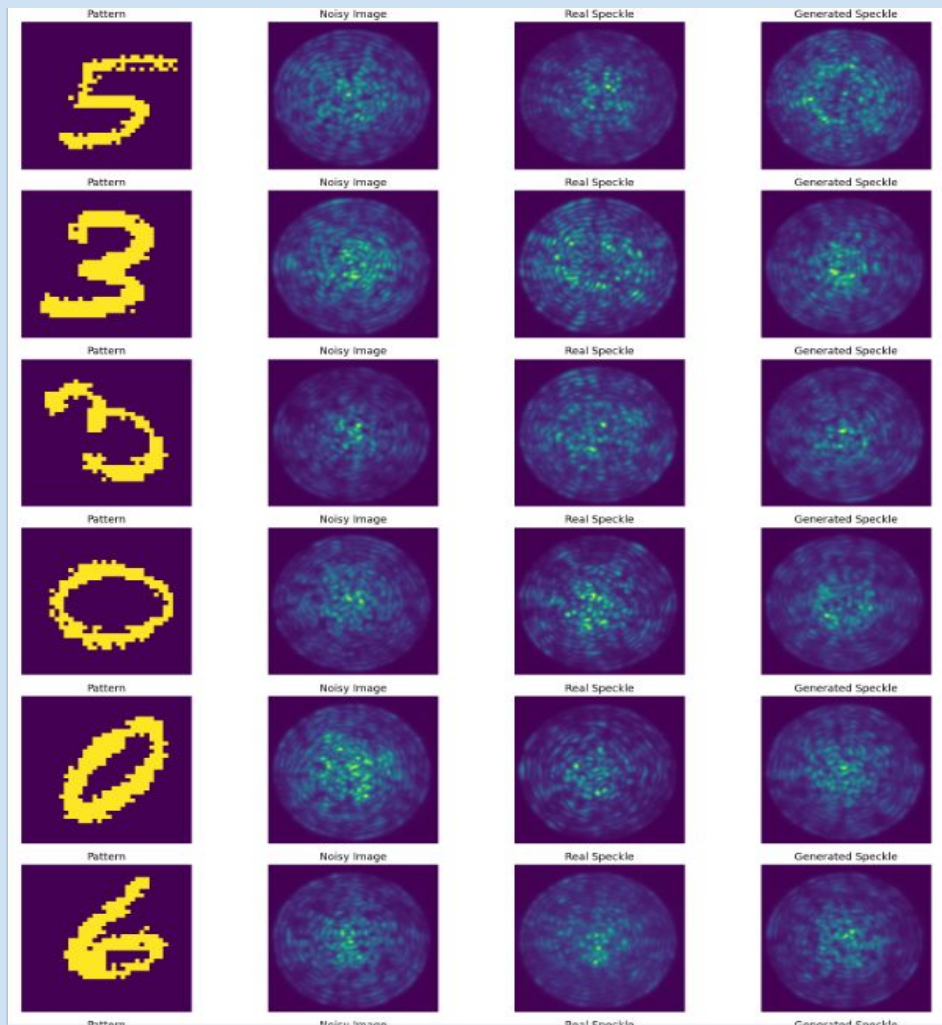
- a) GPU: NVIDIA A100 (QMUL HPC cluster)
- b) Implementation: Python, PyTorch

Training Configuration

- c) Batch size & epochs: 64 & 500
- d) Optimizer & learning rate: Cosine Annealing LR , 1e-4

Final Results

The results of proposed model
with better reconstructions.



THANKYOU